Prabhavit Gupta



Lead Product Engineer: Python, SQL, RAG & AI Data Infrastructure

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (CET Time Zone / EU Permanent Resident)

Summary

Lead Product Engineer and Software Architect with 3 years of commercial experience taking complex data and AI systems from 0 to 1 in production. Proven track record turning proprietary data assets into reliable, searchable intelligence layers combining Python, SQL data infrastructure, Retrieval-Augmented Generation (RAG), and TypeScript services. Experienced owning end-to-end technical delivery, designing clean API contracts, and establishing automated testing and evaluation standards. Based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization, no sponsorship required); available immediately for full-time EU remote roles.

Technical Expertise

- **Data Infrastructure & SQL:** PostgreSQL (relational schema design, complex analytical SQL, JSONB indexing, connection pooling), Redis task queues, ETL data ingestion pipelines.
- **AI Engineering & RAG:** Retrieval-Augmented Generation, vector embeddings, hybrid semantic and keyword search, context injection, LangChain, tool calling, model evaluation harnesses.
- **Backend & Full-Stack Development:** Python (FastAPI, asyncio, pandas, NumPy, Pydantic), TypeScript, Node.js, Next.js, RESTful microservices, structured JSON validation.
- **Architecture & 0-to-1 Ownership:** Technical discovery, modular platform architecture, schema migrations, asynchronous messaging, rate limiting, and observability.
- **DevOps & Testing:** Docker containerization, AWS/GCP cloud services, CI/CD pipeline automation, GitOps, automated regression and contract testing with Playwright.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Conversational). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Led 0-to-1 product engineering of customer-facing AI and data retrieval platforms for a marketplace serving 4,500+ active enterprise users across Europe.
- Architected high-performance Python (FastAPI) and PostgreSQL data pipelines, converting raw multi-source datasets into structured, searchable intelligence assets.
- Implemented production RAG systems with hybrid vector search, dynamic context window management, and automated evaluation harnesses measuring precision and recall.
- Designed asynchronous task processing with Redis, decoupling resource-intensive data indexing from synchronous client API requests.
- Led full-stack TypeScript and Node.js modernization, ensuring high performance, zero downtime, and seamless data synchronization.
- Authored core technical documentation, database schemas, and CI/CD automated test suites with Playwright, ensuring platform stability across rapid release cycles.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and delivery of custom full-stack web platforms and automated data synchronization services.
- Built ETL data ingestion pipelines connecting third-party APIs to PostgreSQL databases, maintaining strict data freshness and provenance.
- Created interactive prototypes and automated validation checks to verify functional specifications prior to production delivery.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Automation & Integration Developer *Venice, Italy*

- Developed custom Python utilities, SQL data models, and LLM integrations for European clients, automating document parsing and classification.
- Built robust data validation routines ensuring incoming unstructured web content conformed to rigorous relational database schemas.
- Authored technical architecture specifications, API documentation, and deployment guides for client engineering teams.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.